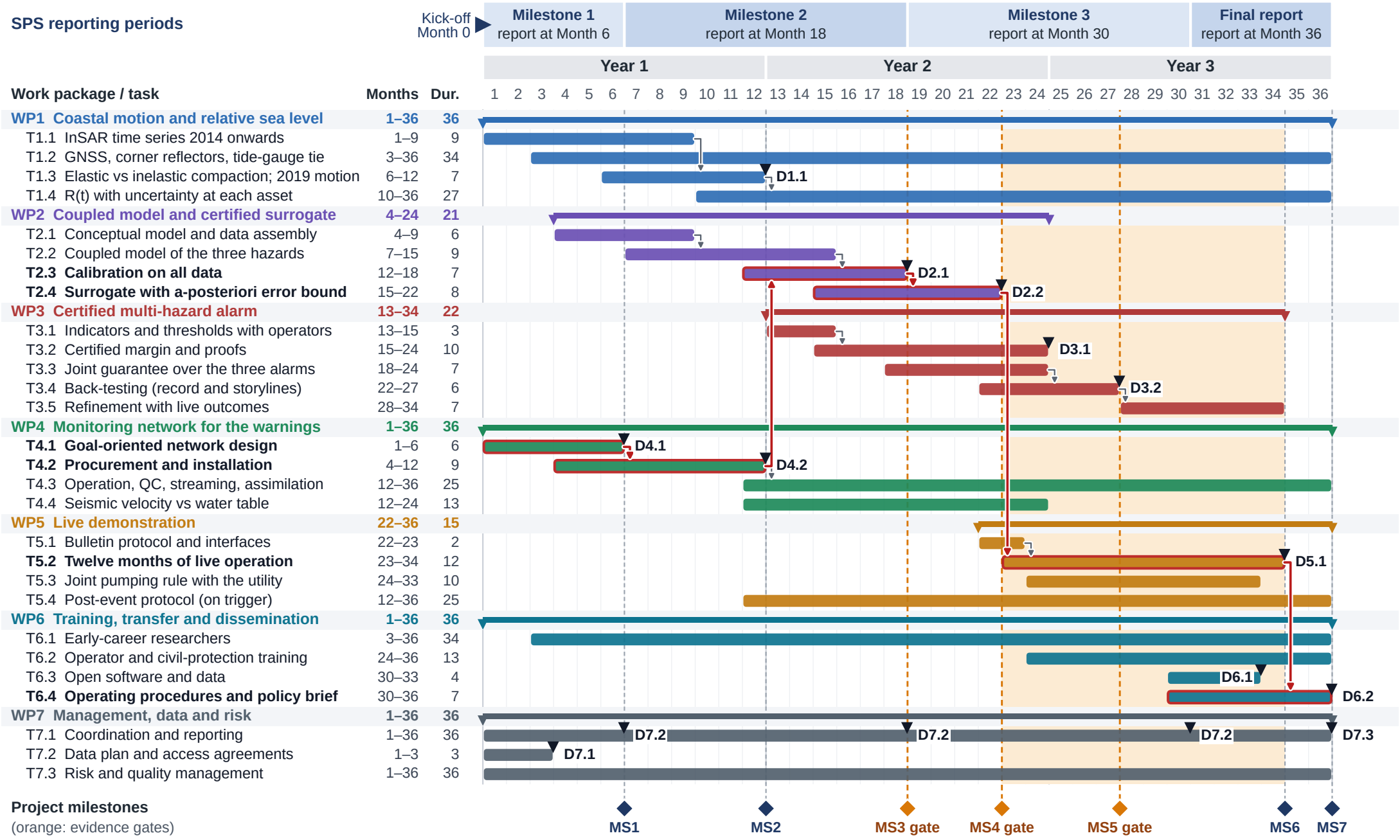
# CERTAIN-COAST — Project schedule (36 months)

Certified early warning to protect water, port and energy infrastructure on a sinking Adriatic coast. Months count from the SPS kick-off meeting (Month 0).




| Work package / task | Months | Dur. |
|---|---|---|
| **WP1 Coastal motion and relative sea level** | **1–36** | **36** |
| T1.1 InSAR time series 2014 onwards | 1–9 | 9 |
| T1.2 GNSS, corner reflectors, tide-gauge tie | 3–36 | 34 |
| T1.3 Elastic vs inelastic compaction; 2019 motion | 6–12 | 7 |
| T1.4 R(t) with uncertainty at each asset | 10–36 | 27 |
| **WP2 Coupled model and certified surrogate** | **4–24** | **21** |
| T2.1 Conceptual model and data assembly | 4–9 | 6 |
| T2.2 Coupled model of the three hazards | 7–15 | 9 |
| **T2.3 Calibration on all data** | 12–18 | 7 |
| **T2.4 Surrogate with a-posteriori error bound** | 15–22 | 8 |
| **WP3 Certified multi-hazard alarm** | **13–34** | **22** |
| T3.1 Indicators and thresholds with operators | 13–15 | 3 |
| T3.2 Certified margin and proofs | 15–24 | 10 |
| T3.3 Joint guarantee over the three alarms | 18–24 | 7 |
| T3.4 Back-testing (record and storylines) | 22–27 | 6 |
| T3.5 Refinement with live outcomes | 28–34 | 7 |
| **WP4 Monitoring network for the warnings** | **1–36** | **36** |
| **T4.1 Goal-oriented network design** | 1–6 | 6 |
| **T4.2 Procurement and installation** | 4–12 | 9 |
| T4.3 Operation, QC, streaming, assimilation | 12–36 | 25 |
| T4.4 Seismic velocity vs water table | 12–24 | 13 |
| **WP5 Live demonstration** | **22–36** | **15** |
| T5.1 Bulletin protocol and interfaces | 22–23 | 2 |
| **T5.2 Twelve months of live operation** | 23–34 | 12 |
| T5.3 Joint pumping rule with the utility | 24–33 | 10 |
| T5.4 Post-event protocol (on trigger) | 12–36 | 25 |
| **WP6 Training, transfer and dissemination** | **1–36** | **36** |
| T6.1 Early-career researchers | 3–36 | 34 |
| T6.2 Operator and civil-protection training | 24–36 | 13 |
| T6.3 Open software and data | 30–33 | 4 |
| **T6.4 Operating procedures and policy brief** | 30–36 | 7 |
| **WP7 Management, data and risk** | **1–36** | **36** |
| T7.1 Coordination and reporting | 1–36 | 36 |
| T7.2 Data plan and access agreements | 1–3 | 3 |
| T7.3 Risk and quality management | 1–36 | 36 |

**Project milestones**
(orange: evidence gates)

Task (colour = work package) · Critical path · Needs (red: on the critical path) · Deliverable · Milestone / gate · Live operation, Months 23–34

*A task "12–18" runs from the start of Month 12 to the end of Month 18. Every dependency, with what it passes on, is listed on the next page; deliverables by SPS milestone and the buffers follow.*

# CERTAIN-COAST — Steps, durations and dependencies

7 work packages, 28 tasks, 36 months. "Needs" lists the tasks each step depends on and what they pass on; a step that needs one still running starts on its preliminary results. Tasks on the critical path, whose chain decides the end date, are in bold.

| Task | Step | Months | Dur. | Needs (what it passes on) | Critical |
|---|---|---|---|---|---|
| WP1 | **Coastal motion and relative sea level** | 1–36 | 36 | | |
| T1.1 | Sentinel-1 InSAR time series (2014 onwards) over the three asset areas | 1–9 | 9 | Starts at kick-off (open Sentinel-1 archive) | |
| T1.2 | GNSS station(s) and corner reflectors at the assets; tie to the tide gauge | 3–36 | 34 | T7.2 (site access) | |
| T1.3 | Elastic and inelastic compaction separated; pumping limit without inelastic compaction; 2019 coseismic and postseismic motion | 6–12 | 7 | T1.1 (InSAR time series) | |
| T1.4 | R(t) series with uncertainty at each asset, updated through the demonstration | 10–36 | 27 | T1.2 (GNSS tie to the tide gauge); T1.3 (compaction separated) | |
| WP2 | **Coupled model and certified surrogate** | 4–24 | 21 | | |
| T2.1 | Conceptual model and data assembly: geology, wells, port and landfall geotechnical data | 4–9 | 6 | T7.2 (operator data) | |
| T2.2 | Coupled model: variable-density flow, salt memory, compaction, FS(x,t) = CRR/CSR(z_w), coastal water level on R(t) | 7–15 | 9 | T2.1 (conceptual model and data) | |
| **T2.3** | **Calibration on heads, conductivity, chloride, InSAR and seismic velocity** | 12–18 | 7 | T2.2 (model); T4.2 (data streaming); T1.3 (land motion) | Yes |
| **T2.4** | **Neural or kernel operator surrogate with a-posteriori error bound** | 15–22 | 8 | T2.3 (calibrated parameters; preliminary ones first) | Yes |
| WP3 | **Certified multi-hazard alarm** | 13–34 | 22 | | |
| T3.1 | Hazard indicators and thresholds agreed with the operators (chloride limit, FS < 1, design water level) | 13–15 | 3 | T2.2 (model outputs that define the indicators) | |
| T3.2 | Margin construction and proofs: error bound + online conformal risk control + extreme-aware term | 15–24 | 10 | T3.1 (thresholds); T2.4 (surrogate error bound) | |
| T3.3 | Joint guarantee over the three alarms; tolerances $\alpha_\kappa$ allocated by consequence | 18–24 | 7 | T3.2 (margin of each alarm) | |
| T3.4 | Back-testing on the historical record and on storylines beyond it | 22–27 | 6 | T3.3 (joint guarantee); T2.4 (storylines from the surrogate) | |
| T3.5 | Alarm refinement with the outcomes of the live operation | 28–34 | 7 | T3.4 (back-testing, MS5); T5.2 (live outcomes) | |
| WP4 | **Monitoring network for the warnings** | 1–36 | 36 | | |
| **T4.1** | **Goal-oriented design across sensor types: piezometers with conductivity, chloride sampling, GNSS, seismic noise, nodes, fibre** | 1–6 | 6 | Starts at kick-off (existing and remote-sensing data) | Yes |
| **T4.2** | **Procurement and installation** | 4–12 | 9 | T4.1 (design; sensors sure to be needed ordered in Month 4) | Yes |
| T4.3 | Operation, quality control, data streaming and assimilation | 12–36 | 25 | T4.2 (installed network) | |
| T4.4 | Seismic velocity calibrated against the water table at the port or landfall | 12–24 | 13 | T4.2 (installed seismic sensors and piezometers) | |
| WP5 | **Live demonstration** | 22–36 | 15 | | |
| T5.1 | Bulletin protocol and decision interfaces with the operators and civil protection | 22–23 | 2 | T3.1 (thresholds); T3.2 (alarm margins) | |
| **T5.2** | **Twelve months of live operation, bulletins logged and outcomes verified** | 23–34 | 12 | T2.4 (surrogate passes gate MS4); T5.1 (bulletin protocol) | Yes |
| T5.3 | Pumping rule evaluated against the three hazards together and agreed with the water utility | 24–33 | 10 | T2.4 (surrogate); T5.2 (live operation) | |
| T5.4 | Post-event protocol after strong earthquakes or surges (on trigger) | 12–36 | 25 | T4.2 (network installed) | |
| WP6 | **Training, transfer and dissemination** | 1–36 | 36 | | |
| T6.1 | Early-career researchers: exchanges, schools and supervised work in WP1–WP5 | 3–36 | 34 | Starts with recruitment; supervised work in WP1–WP5 | |
| T6.2 | Training of operator and civil-protection staff | 24–36 | 13 | T5.1 (bulletin protocol); T5.2 (live operation) | |
| T6.3 | Open software and data release with documentation | 30–33 | 4 | T3.4 (back-tested algorithms); T2.4 (surrogate) | |
| **T6.4** | **Operating procedures and policy brief** | 30–36 | 7 | T5.2 (live operation) | Yes |
| WP7 | **Management, data and risk** | 1–36 | 36 | | |
| T7.1 | Coordination, steering committee, reporting | 1–36 | 36 | Runs throughout | |
| T7.2 | Data management and access agreements with the water utility, port and landfall operators | 1–3 | 3 | Starts at kick-off | |
| T7.3 | Risk and quality management | 1–36 | 36 | Runs throughout | |

## Milestones and deliverables by SPS reporting milestone

The SPS reporting milestones are those of the application form and of the Info tab of the budget workbook. Each report gives the project milestones reached in its period, with the result of their pass tests, and the deliverables due in it.

| SPS milestone | Project milestones reached, with their pass tests | Deliverables due |
|---|---|---|
| **Kick-off**, Month 0 | Project start; steering committee constituted; access agreements started (T7.2) | — |
| **Milestone 1**, Month 6<br>First report on Months 1–6 | **MS1** (M6) Network design agreed. Pass: D4.1 approved by the steering committee and the operators | D7.1 (M3) Data management plan and access agreements<br>D4.1 (M6) Network design report<br>D7.2 (M6) Progress report to SPS |
| **Milestone 2**, Month 18<br>Second report on Months 7–18 | **MS2** (M12) Network streaming; coastal motion known. Pass: D4.2 (≥ 90% data availability over the first month) and D1.1 delivered<br>**MS3** (M18) Coupled model calibrated (evidence gate). Pass: D2.1: held-out observations reproduced within their stated uncertainty | D1.1 (M12) Coastal motion atlas and R(t) series<br>D4.2 (M12) Network installed and streaming<br>D2.1 (M18) Calibrated coupled model<br>D7.2 (M18) Progress report to SPS |
| **Milestone 3**, Month 30<br>Third report on Months 19–30 | **MS4** (M22) Go/no-go for live operation (evidence gate). Pass: D2.2 surrogate error bound met on the test set; alarm prototype passes a dry run<br>**MS5** (M27) Miss-rate targets confirmed in back-testing (evidence gate). Pass: D3.2: each miss rate ≤ $\alpha_k$ with its confidence interval reported | D2.2 (M22) Certified surrogate<br>D3.1 (M24) Certified alarm algorithms and proofs<br>D3.2 (M27) Back-testing report<br>D7.2 (M30) Progress report to SPS |
| **Final report**, Month 36<br>Final report on Months 31–36 | **MS6** (M34) Twelve months of live operation completed. Pass: D5.1 with the bulletin log and verified outcomes<br>**MS7** (M36) Hand-over to the operators. Pass: D6.2 procedures adopted; operator staff run the system unaided | D6.1 (M33) Software and data release<br>D5.1 (M34) Live demonstration report<br>D6.2 (M36) Operating procedures and policy brief<br>D7.3 (M36) Final report |

## Critical path and buffers

| Step on the critical path | Months | Ends at | Buffer |
|---|---|---|---|
| Network design (T4.1) | M1–6 | MS1 | Equipment ordered in Month 4, before the design is final, for sensors certain to be needed |
| Procurement and installation (T4.2) | M4–12 | MS2 | Installation window of 8 months; highest-value sensors installed first |
| Calibration of the coupled model (T2.3) | M12–18 | MS3 (gate) | Model built in stages from Month 7, so calibration starts as soon as data stream |
| Certified surrogate (T2.4) | M15–22 | MS4 (gate) | Starts before calibration ends, on preliminary parameters |
| Live operation (T5.2) | M23–34 | MS6 | Back-testing (to Month 27) runs in parallel, not before |
| Hand-over (T6.4) | M30–36 | MS7 | Operators issue all bulletins from Month 34; procedures drafted from Month 30 |

A slip of up to two months at the go/no-go gate (MS4) is absorbed: live operation then runs from Month 25 to Month 36, still twelve months, and the operators still issue the bulletins in the final months. A longer slip triggers the fallbacks of the Risk Assessment (R5, R6) and is decided by the steering committee.

## What is new in this schedule

- **Every SPS report carries a decision, not only progress.** The reports fall on the evidence: Month 6 on the agreed network design (MS1), Month 18 on the calibration gate (MS3), Month 30 after the go/no-go and miss-rate gates (MS4, MS5), Month 36 on the hand-over (MS7). NATO sees each pass test and its result when it is decided.
- **Gates, not only dates.** Live operation starts only after the coupled model passes its validation test (MS3) and the surrogate meets its error bound (MS4); the certified miss rates are confirmed in back-testing (MS5). Each gate has a pass test and a fallback fixed in advance.
- **A full year live, with slack.** Back-testing runs alongside live operation, so twelve full months of operation fit even after a two-month slip at MS4, and they include every season of the hazards.
- **The hand-over is inside the schedule.** The operators issue every bulletin in Months 34–36, so the end state is demonstrated before the grant ends.